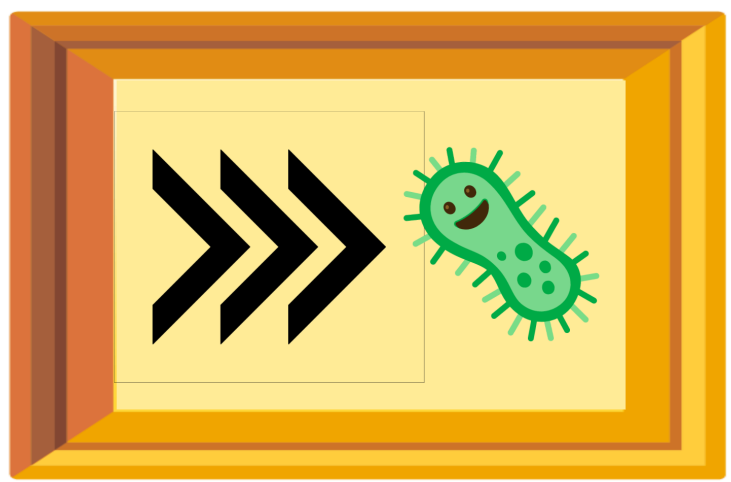




OME-Arrow: Unifying Images, Metadata, and Features in an Interoperable Data Model

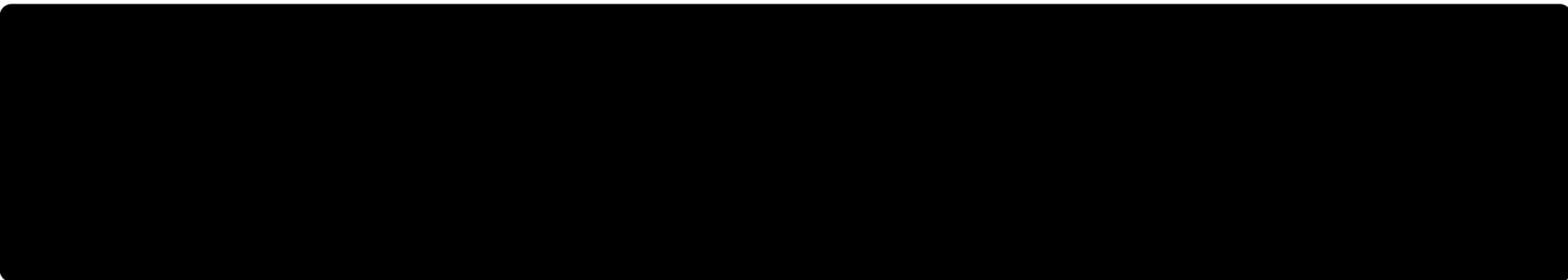
Dave Bunten¹, Jenna Tomkinson¹, Michael Lippincott¹, Cameron Mattson¹, Julia B. Curd¹, Gregory P. Way¹

¹Department of Biomedical Informatics, University of Colorado Anschutz Medical Campus



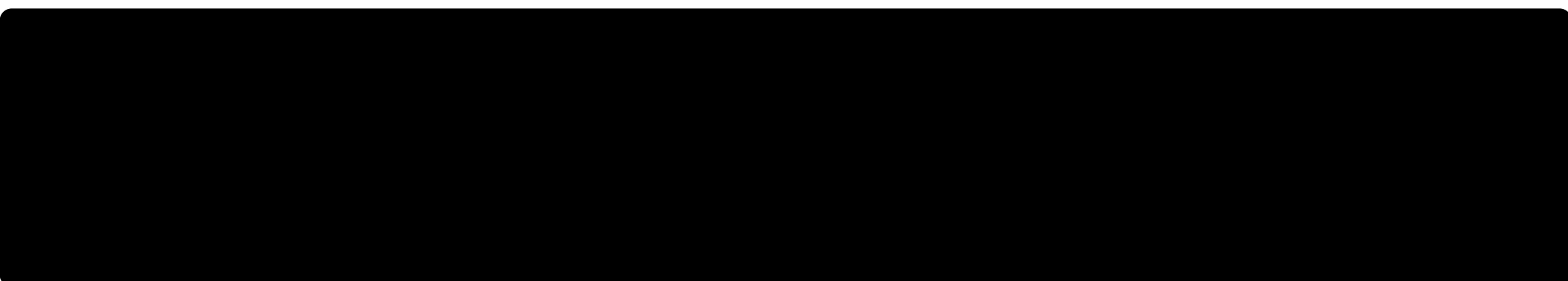
Anschutz

I. Why OME-Arrow?



Modern bioimaging workflows depend on connecting **images**, **metadata**, and **derived measurements**. When these are split across disconnected files and systems, analysis is harder to join, reproduce, and scale. **OME-Arrow** provides a linked, queryable data model for these components in code-first and SQL-first analytical workflows.

II. An interoperable image data model



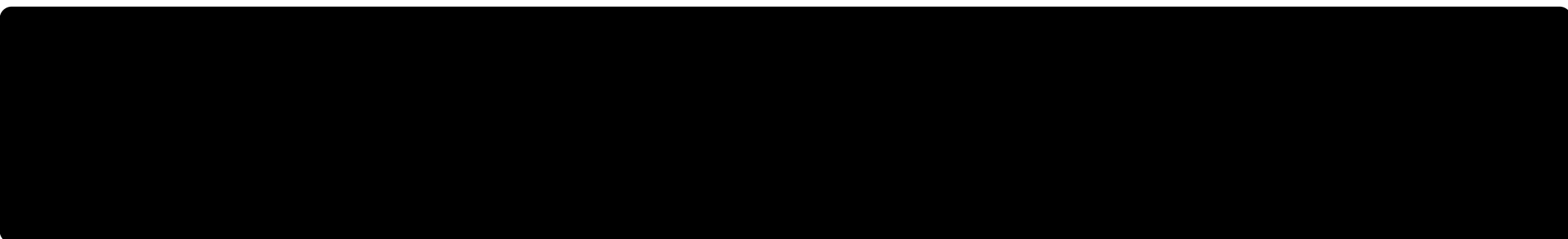
By representing image content as **Arrow-compatible structures** alongside metadata and features, OME-Arrow enables:

- **Explicit relationships** between pixels, metadata, and derived features
- **Direct joins and filtering** in SQL and DataFrame workflows
- **Cross-language interoperability** through Arrow-native representations
- **Standards alignment** with the broader open bioimaging ecosystem

III. How OME-Zarr integrates

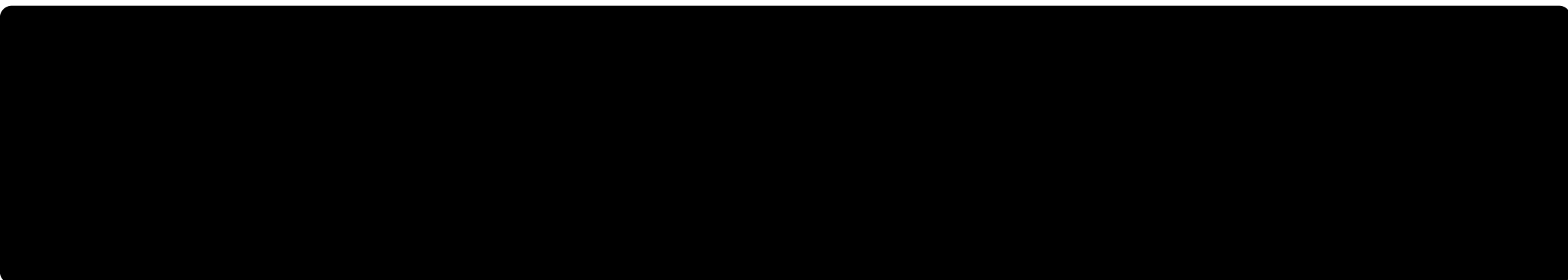
OME-Zarr is excellent for cloud-native image storage and visualization-oriented distribution. OME-Arrow is complementary, not a replacement. It is designed for cases where images must be queried together with tabular metadata and measurements in one analytical system. In many projects, both are useful and intentionally used together:

- **OME-Zarr / OME-TIFF** for image exchange and ecosystem compatibility
- **OME-Arrow / Parquet / Arrow-native tables** for joins, filtering, and analytics at scale
- **Lance** appears to be a promising OME-Arrow table backend for large-image random access in tabular workflows



IV. Benchmark signal (preliminary)

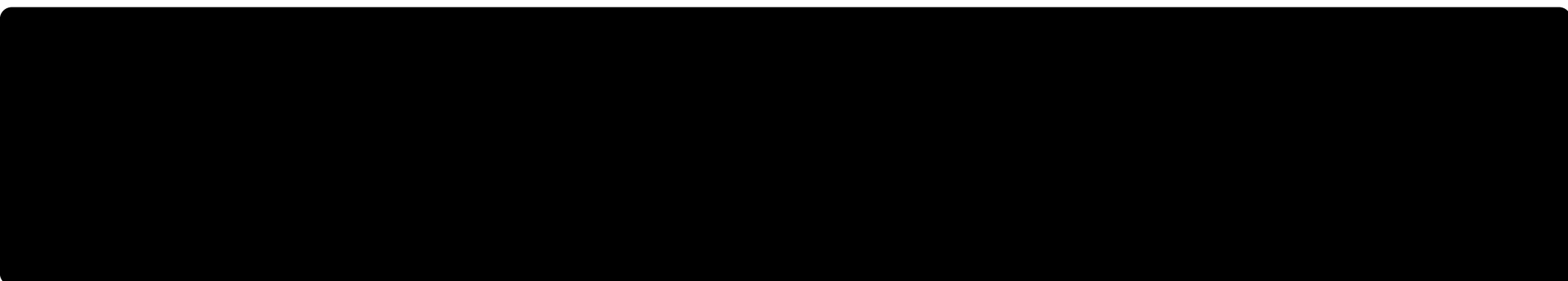
Current benchmark results suggest operation-dependent tradeoffs rather than one universal winner. Directory-per-image OME-Zarr can perform strongly for sparse random image access patterns. Arrow-table-native layouts can provide practical advantages for table-centric join/filter workloads and broader analytical interoperability. Lance shows competitive random-access behavior in OME-Arrow-oriented tests, making it a strong candidate for large image-linked table repositories.



V. Supported workflows

OME-Arrow supports practical data movement across common microscopy formats and downstream analysis environments:

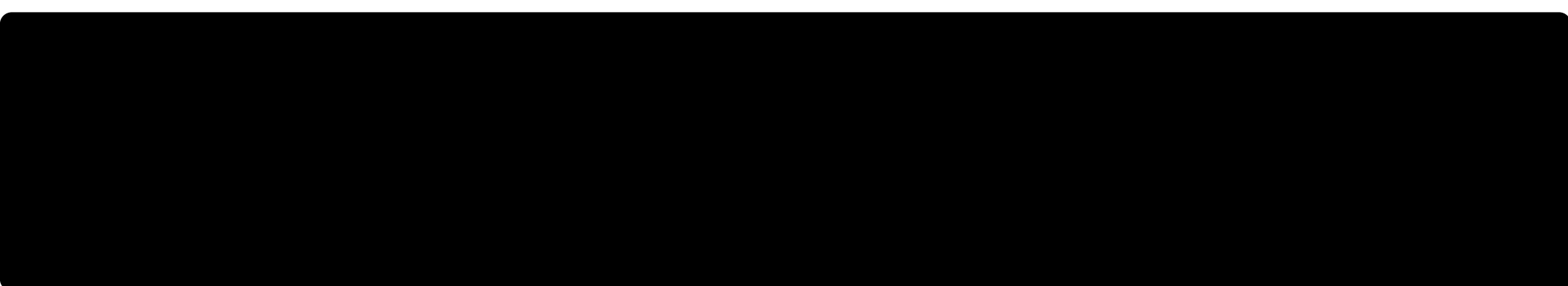
- **Ingest** from TIFF, OME-Zarr, and NumPy arrays
- **Export** to OME-Parquet, OME-Zarr, OME-TIFF, Vortex, and Lance-compatible table workflows
- **Standalone/local analysis** with scripts, SQL engines, and DuckDB
- **Lazy scan-style access** for large datasets
- **Tensor pathways** for machine learning and zero-copy style workflows
- **Visualization** through napari-ome-arrow
- **Feature-centric workflows** through CytoDataFrame



VI. Quick technical demonstration

```
from ome_arrow import OMEArrow

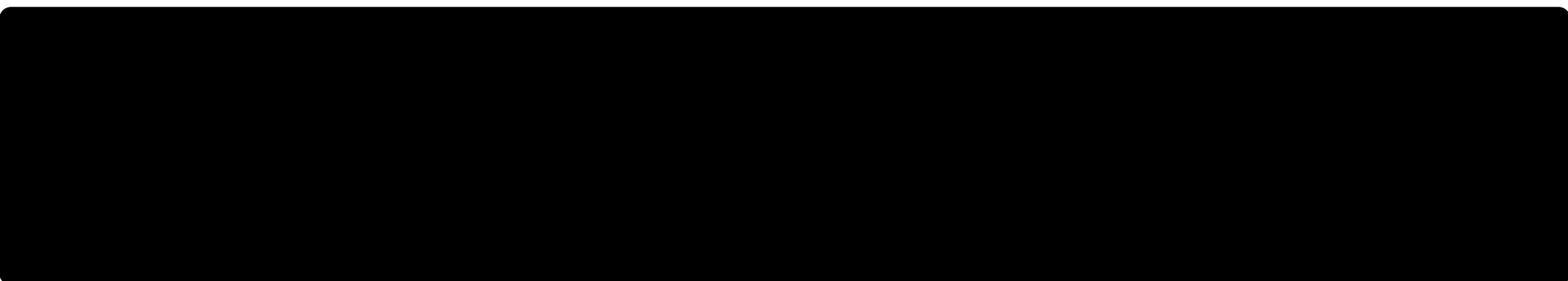
oa = OMEArrow("image.ome.zarr") # ingest common formats
oa.export(how="ome-parquet", out="image.parquet")
lazy = OMEArrow.scan("image.parquet").slice_lazy(0, 512, 0, 512).collect()
```



VII. Integration with the ecosystem

OME-Arrow is intentionally modular and complements other open-source bioimaging tools rather than replacing them.

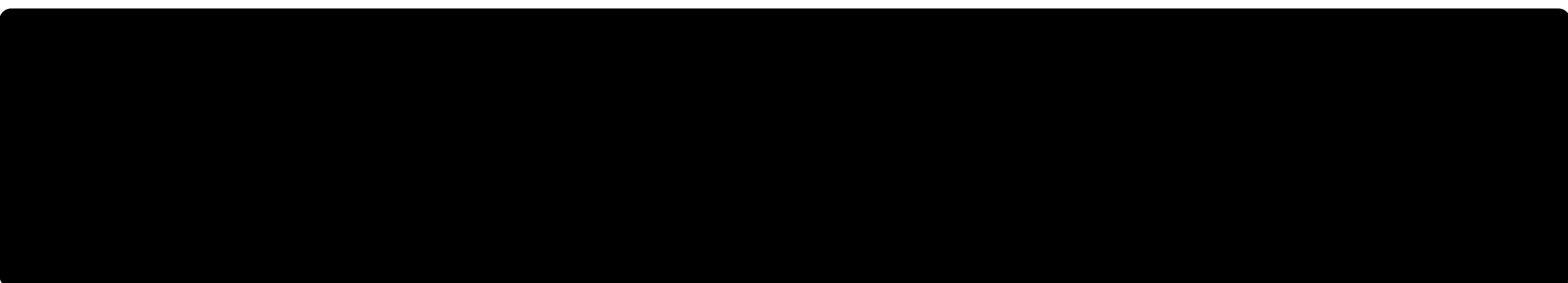
- **napari-ome-arrow** enables interactive visualization of OME-Arrow and OME-Parquet data in napari.
- **CytoDataFrame** is the practical analysis layer that brings image-linked OME-Parquet data into DataFrame-centered notebook workflows for image-based profiling.
- **coSMicQC** can operate on feature-rich datasets that are paired with image context through Cytomining ecosystem tools.
- **iceberg-bioimage** helps connect existing formats and is defining an Iceberg-oriented warehouse specification for how Cytomining tools bring images, features, and metadata together with versioning and schema evolution.
- **Lance** appears to be a suitable tabular option for large-image access within table-native workflows, with competitive random-access behavior in preliminary OME-Arrow benchmarks.



VIII. In practice: ALSF pediatric cancer workflows

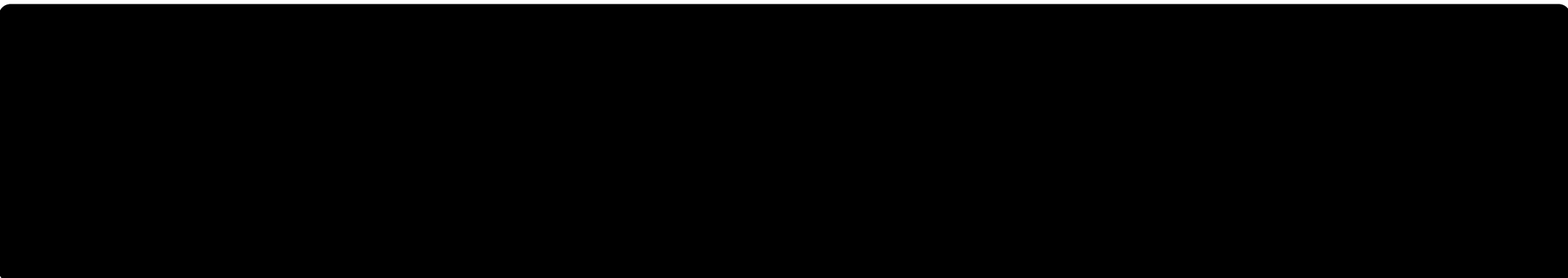
This framework is being used with **Alex's Lemonade Stand Foundation (ALSF)** data to support integrated image-based profiling. Instead of keeping raw images, extracted features, and experimental metadata in disconnected stores, OME-Arrow enables a linked analytical representation that can be queried and validated as one system.

- **Faster hypothesis iteration:** join cellular phenotypes to treatment and sample context without custom file stitching.
- **Improved reproducibility:** preserve explicit data relationships across preprocessing, QC, and downstream modeling.
- **Scalable operations:** run the same model from local notebooks to larger repository/warehouse settings.
- **Cytomining alignment:** OME-Arrow + CytoDataFrame + coSMicQC provide a coherent path from image ingest through feature analysis and quality control.



IX. Why this matters

OME-Arrow offers a **standards-aligned**, **queryable**, and **language-agnostic** way to keep imaging data connected to the metadata and measurements that scientists actually analyze. This makes it easier to move between image storage, tabular analytics, interactive visualization, and machine learning without inventing a new data model for each tool. In image-based profiling contexts, this supports end-to-end analysis where raw images, extracted features, and experimental context remain connected from local workflows to large data repositories. For skeptical users, the key point is practical: OME-Arrow does not ask teams to abandon OME-Zarr or OME-TIFF; it provides the missing analytical layer that makes those assets easier to query, relate, and operationalize at scale.



X. Acknowledgements

We thank those who have inspired, contributed, or helped support OME-Arrow and the broader open bioimaging ecosystem:

- Open source science from the **Open Microscopy Environment**
- The communities behind **Apache Arrow**, **napari**, and Cytomining ecosystem projects
- Members of the **Way Lab** at the University of Colorado Anschutz Medical Campus
- **Department of Biomedical Informatics** within the School of Medicine at the University of Colorado Anschutz Medical Campus